Temperature Monitoring

System Using Arduino

Simple Project Documentation

Project Title	Temperature Monitoring System Using Arduino
Components	Arduino Uno, DHT11, LCD 16x2 with I2C Module
Language	Arduino C++
Date	June 08, 2026
Status	Completed

1. Introduction

A Temperature Monitoring System is a simple yet practical embedded project that measures and displays the surrounding temperature and humidity in real time. In this project, an Arduino Uno microcontroller is interfaced with a DHT11 temperature and humidity sensor and a 16x2 LCD display connected via an I2C module.

The DHT11 sensor detects environmental conditions and sends digital data to the Arduino. The Arduino processes the data and shows the temperature (in Celsius) and humidity (in %) on the LCD screen. The readings are also sent to the Arduino IDE Serial Monitor for live observation. This project is ideal for beginners to learn sensor interfacing, I2C communication, and LCD display control.

2. Objective

The main objectives of this project are:

- To interface the DHT11 sensor with Arduino Uno for reading temperature and humidity
- To display temperature (°C) and humidity (%) on a 16x2 LCD with I2C module
- To show sensor readings on the Arduino IDE Serial Monitor at 9600 baud
- To handle sensor errors gracefully and display error messages on LCD
- To understand I2C communication between Arduino and the LCD module
- To build a beginner-friendly embedded project using Arduino IDE libraries

3. Components Required

#	Component	Specification	Quantity
1	Arduino Uno	ATmega328P, 5V, 14 digital I/O pins	1
2	DHT11 Sensor	Temperature: 0–50°C, Humidity: 20–90% RH	1
3	LCD 16x2 Display	16 columns x 2 rows character LCD	1
4	I2C LCD Module (PCF8574)	I2C address 0x27, reduces wiring to 2 pins	1
5	Jumper Wires	Male-to-male and male-to-female	10–15
6	Breadboard	830-tie standard breadboard	1

7	USB Cable	Type-A to Type-B for Arduino power/upload	1
8	PC / Laptop	For Arduino IDE and Serial Monitor	1

4. Working Principle

Step 1 — Sensor Reads Data:

The DHT11 sensor measures surrounding temperature and humidity. It sends a digital signal to Arduino digital pin 7. The signal encodes both temperature (in °C) and humidity (in %) in a single data line using a custom 1-wire protocol.

Step 2 — Arduino Processes Data:

The Arduino uses the DHT library to decode the signal from the DHT11 sensor. It reads three values: temperature in Celsius (tempC), temperature in Fahrenheit (tempF), and relative humidity (humidity). If the values are invalid (isnan check), an error message is shown.

Step 3 — LCD Displays Output:

The processed values are sent to the 16x2 LCD over I2C communication (only 2 wires: SDA and SCL). Line 1 of the LCD shows the temperature in °C with the degree symbol. Line 2 shows the humidity percentage. The LCD refreshes every 2 seconds.

Step 4 — Serial Monitor Output:

At the same time, all three readings (tempC, tempF, humidity) are printed to the Arduino IDE Serial Monitor at 9600 baud, which is useful for debugging and logging data on a computer.

5. Data Flow Diagram (DFD)

The diagram below shows how data flows from the DHT11 sensor through the Arduino Uno to the LCD and Serial Monitor:

DHT11 Sensor	→ data	**Arduino Uno**	→ I2C	**LCD 16x2 (I2C)**	
		↓			
		Serial Monitor			

Description of data flow:

- DHT11 Sensor → Arduino Uno : Temperature & humidity via digital pin 7 (1-wire protocol)
- Arduino Uno → LCD 16x2 : Processed readings sent over I2C (SDA pin A4, SCL pin A5)
- Arduino Uno → Serial Monitor : Raw values printed at 9600 baud via USB

6. Circuit Connections

DHT11 Sensor to Arduino:

DHT11 Pin	Arduino Pin	Description
VCC	5V	Power supply
GND	GND	Ground
DATA	Pin 7	Digital data signal (DHTPIN)

I2C LCD Module to Arduino:

I2C Module Pin	Arduino Pin	Description
VCC	5V	Power supply
GND	GND	Ground
SDA	A4	I2C data line
SCL	A5	I2C clock line

7. Program

Upload the following code to the Arduino Uno using the Arduino IDE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

#define DHTPIN  7          // DHT11 connected to Digital Pin 7
#define DHTTYPE DHT11      // Sensor type: DHT11

DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);  // I2C address 0x27, 16 cols, 2 rows

void setup() {
  Serial.begin(9600);
  dht.begin();
  lcd.init();
  lcd.backlight();

  // Welcome message
  lcd.setCursor(0, 0);
  lcd.print(" Temp Monitor ");
  lcd.setCursor(0, 1);
  lcd.print(" Initializing..");
  delay(2000);
  lcd.clear();
}

void loop() {
  delay(2000);   // DHT11 needs 2 sec between readings

  float humidity = dht.readHumidity();
  float tempC    = dht.readTemperature();
  float tempF    = dht.readTemperature(true);

  // Check for sensor read error
  if (isnan(humidity) || isnan(tempC)) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Sensor Error!");
    lcd.setCursor(0, 1);
    lcd.print("Check wiring...");
    Serial.println("ERROR: Failed to read from DHT sensor!");
    return;
  }

  // Line 1: Temperature
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(tempC, 1);       // 1 decimal place
  lcd.print((char)223);      // Degree symbol °
```

```
  lcd.print("C");

  // Line 2: Humidity
  lcd.setCursor(0, 1);
  lcd.print("Humidity: ");
  lcd.print(humidity, 1);
  lcd.print("%");

  // Print to Serial Monitor
  Serial.print("Temperature: ");
  Serial.print(tempC);
  Serial.print("\u00b0C | ");
  Serial.print(tempF);
  Serial.print("\u00b0F | Humidity: ");
  Serial.print(humidity);
  Serial.println("%");
}
```

Library Installation (Arduino IDE):

- DHT sensor library : Sketch > Include Library > Manage Libraries > search "DHT sensor library" by Adafruit
- LiquidCrystal I2C : search "LiquidCrystal I2C" by Frank de Brabander
- Wire.h : Built-in with Arduino IDE, no installation needed

8. Output

LCD Display Output:

On power-on, the LCD shows a welcome message for 2 seconds, then starts showing live readings:

Welcome screen (2 seconds):

```
 Temp Monitor
 Initializing..
```

Normal reading screen (updates every 2 seconds):

```
Temp: 28.5 C
Humidity: 65.0%
```

Sensor error screen:

```
Sensor Error!
Check wiring...
```

Serial Monitor Output (9600 baud):

```
Temperature: 28.50°C | 83.30°F | Humidity: 65.00%
Temperature: 28.60°C | 83.48°F | Humidity: 64.80%
Temperature: 28.50°C | 83.30°F | Humidity: 65.10%
ERROR: Failed to read from DHT sensor!
```

9. Real-World Applications

Application Area	Description
Home Automation	Monitor room temperature and humidity for smart AC or fan control systems.
Greenhouse Farming	Track temperature and humidity inside greenhouses to maintain ideal crop growth conditions.

Cold Chain Logistics	Monitor temperature of refrigerated trucks and storage units to ensure product safety.
Server Rooms	Alert administrators when server room temperature exceeds safe operating limits.
Hospitals & Labs	Ensure sterile environments maintain precise temperature and humidity for medical equipment.
Weather Stations	Build low-cost personal weather stations to log local environmental conditions.
Schools & Colleges	Use as a learning project to teach sensor interfacing, I2C, and Arduino programming.
HVAC Systems	Provide feedback data to heating and cooling systems for efficient energy management.

Conclusion

This project successfully demonstrates how to build a real-time temperature and humidity monitoring system using Arduino Uno, a DHT11 sensor, and a 16x2 LCD with I2C module. The system reads sensor data every 2 seconds, displays it on the LCD screen, and also outputs it to the Serial Monitor for live monitoring.

The project introduces key embedded system concepts such as digital sensor communication, I2C protocol, library usage in Arduino IDE, and error handling. It can be extended further by adding data logging to an SD card, wireless transmission via Wi-Fi or Bluetooth, or threshold-based alerts using a buzzer or LED.